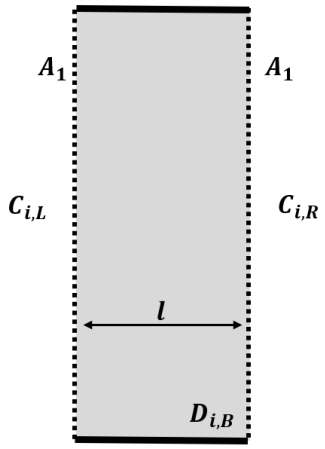
# CBE 150B F2026
# Problem Set 2 - Key

Assigned: September 2, 2026
Due: Septermber 9, 2026

## Problem 1 (X pts)

Consider steady-state mass transport through a slab-shaped membrane of thickness $l$. This membrane separates two binary mixtures of component $i$ in an excess of $B$. The left side is at a higher concentration ($C_{i,L}$) and the right side is at a lower concentration ($C_{i,R}$).




Assume there is no partitioning of $i$ in the membrane. This means the concentration if $i$ inside the surface of the membrane directly contacting the fluid phase is equal to the concentration of the fluid it is contacting.

**a)** Starting from a material balance, determine and sketch the concentration profile of $i$ given a diffusivity of $D_{i,B}$ within the membrane.

Boundary Conditions:
$(1): C_i(z) = C_{i,L}$ at $z = 0$
$(2): C_i(z) = C_{i,R}$ at $z = l$

Perform Material Balance on a shell of thickness $\Delta Z$:
$In - Out + Generation = Accumulation$
Recognize Simplifications to Material Balance:
Steady State: $\frac{dC_i}{dt} = Accumulation = 0$
No Bulk Reaction Occurs: $R_i = Generation = 0$

Shell Balance:

$(N_{i,z} \cdot A)|_z \ - (N_{i,z} \cdot A)|_{z+\Delta z} = 0$

$A$ is constant throughout the slab, so it can $(N_{i,z})|_z \ - (N_{i,z})|_{z+\Delta z} = 0$

Divide the entire shell balance by $\Delta z$.

$\frac{(N_{i,z})|_z \ -(N_{i,z})|_{z+\Delta z}}{\Delta z} = 0$

Apply Definition of an Derivative

$\frac{dN_{i,z}}{dz} = 0$

Expression for Flux of $i$ in a binary system:

$N_{i,z} = -D_{i,B}\frac{dC_i}{dz} + x_i(N_{i,z} + N_{B,z})$

Simplify for Dilute Condition: $x_i(N_{i,z} + N_{B,z}) = 0$ Because $x_i$ is Very Small.

Flux Expression Simplifies to Diffusive Flux Contribution:

$N_{i,z} = -D_{i,B}\frac{dC_i}{dz}$

Apply Flux Expression to the Material Balance:

$\frac{d(-D_{i,B}\frac{dC_i}{dz})}{dz} = 0$

Pull $-D_{i,B}$ out because it is a constant:

$\frac{d^2C_i}{dz^2} = 0$

Separate and Integrate Twice:

$C_i = C_1 z + C_2$

Apply Boundary Conditions:

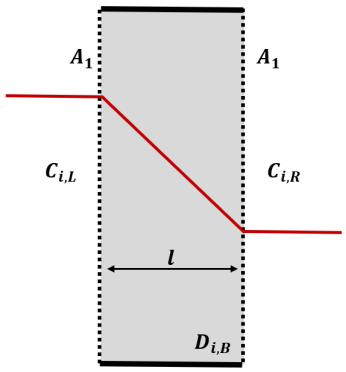
(1) $C_i(z = 0) = C_{i,L}$ and thus $C_2 = C_{i,L}$

(2) $C_i(z = l) = C_{i,R}$ and thus $C_1 = \frac{C_{i,R} - C_{i,L}}{l}$

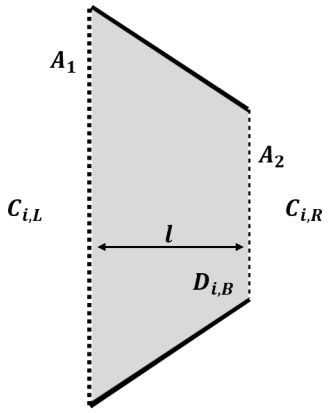
Therefore, the final concentration profile is linear, adhering to the following profile:

$$\boxed{C_i(z) = (C_{i,R} - C_{i,L})\frac{z}{l} + C_{i,L}}$$

The profile will appear as a straight line between the higher concentration on the left and lower concentration on the right.




Now, consider the same physical scenario as in part (a), except the membrane is no longer a slab. Instead it has a trapezoidal cross-section, as seen below.




**b)** Starting again from a material balance, determine the new concentration profile of $i$ within the trapezoidal membrane.

In the case of a slab with a non-constant trapezoidal area, we will begin our solution with the same Boundary Conditions and Shell Balance:

Boundary Conditions:
$(1) : C_i(z) = C_{i,L}$ at $z = 0$
$(2) : C_i(z) = C_{i,R}$ at $z = l$

Perform Material Balance on a shell of thickness $\Delta Z$:
$In - Out + Generation = Accumulation$
Recognize Simplifications to Material Balance:
Steady State: $\frac{dC_i}{dt} = Accumulation = 0$
No Bulk Reaction Occurs: $R_i = Generation = 0$

Shell Balance:
$(N_{i,z} \cdot A)|_z \ - (N_{i,z} \cdot A)|_{z+\Delta z} = 0$

Further, divide this shell balance by $\Delta z$ and use the definition of a derivative. The result identifies that $N_{i,z} \cdot A(z)$ is constant across the membrane.
$\frac{(N_{i,z} \cdot A)|_z \ - (N_{i,z} \cdot A)|_{z+\Delta z}}{\Delta z} = 0$
Applying the definition of a derivative:
$\frac{dN_{i,z} \cdot A}{dz} = 0$

Now, our solution method will change. Unlike the previous section, we can't factor $A$ out of the derivative, because $A$ is no longer constant. Therefore, before continuing, we must find $A(z)$.

Given the boundary conditions that the left side ($z = 0$) has a cross-sectional area of $A_1$, and the right side ($z = l$) has a cross-sectional area of $A_2$, the following expression for $A(z)$ can be derived:
$A(z) = A_1 + \frac{A_2 - A_1}{l} z$

Now, as in part (a), we must use the definition of $N_{i,z}$ to identify how concentration gradients and flux are related.
See Above, same reasoning applies: $N_{i,z} = -D_{i,B} \frac{dC_i}{dz}$
Returning to equation above ($\frac{dN_{i,z} \cdot A}{dz} = 0$), insert the definitions of flux and Area above:
$\frac{d}{dz}((-D_{i,B} \frac{dC_i}{dz}) \cdot (A_1 + \frac{A_2 - A_1}{l} z)) = 0$

$-D_{i,B}$ can be factored out because it is constant throughout the membrane:
$\frac{d}{dz}((\frac{dC_i}{dz}) \cdot (A_1 + \frac{A_2 - A_1}{l} z)) = 0$

Separating the above expression and integrating once:
$(\frac{dC_i}{dz}) \cdot (A_1 + \frac{A_2 - A_1}{l} z) = C_1$

Dividing each side by the expression for area, and separating the concentration gradient yields the following:
$dC_i = \frac{C_1}{(A_1 + \frac{A_2 - A_1}{l} z)} dz$

Integrating the above expression over appropriate bounds (using B.C. (1)):
$\int_{C_{i,L}}^{C_i(z)} dC_i = \int_0^z \frac{C_1}{(A_1 + \frac{A_2 - A_1}{l} z)} dz$

The result is the following:
$(C_i(z) - C_{i,L}) = \frac{C_1 l}{A_2 - A_1} (ln(A_1 + \frac{A_2 - A_1}{l} z) - ln(A_1))$

To solve for $C_1$, we will integrate again between the two boundary conditions:
$\int_{C_{i,L}}^{C_{i,R}} dC_i = \int_0^l \frac{C_1}{(A_1 + \frac{A_2 - A_1}{l} z)} dz$

See the following: $(C_{i,R} - C_{i,L}) = \frac{C_1 l}{A_2 - A_1} (ln(A_2) - ln(A_1))$

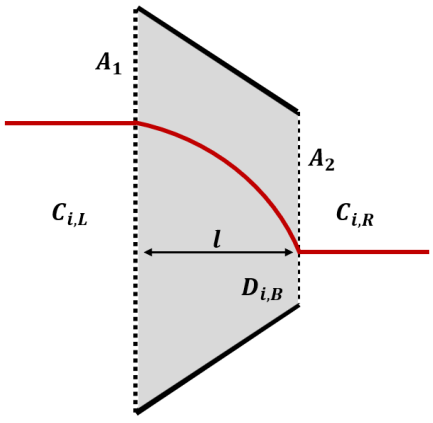
Solving for $C_1$:
$C_1 = \frac{(C_{i,R} - C_{i,L})(A_2 - A_1)}{l(ln(A_2) - ln(A_1))}$

Apply solution for $C_1$ above to the integral solution using only Boundary Condition (1):
$(C_i(z) - C_{i,L}) = (\frac{(C_{i,R} - C_{i,L})(A_2 - A_1)}{l(ln(A_2) - ln(A_1))}) \frac{l}{A_2 - A_1} (ln(A_1 + \frac{A_2 - A_1}{l} z) - ln(A_1))$

Simplifying:

$$\boxed{C_i(z) = \frac{(C_{i,R} - C_{i,L})}{(ln(\frac{A_2}{A_1}))} ln(1 + (\frac{A_2}{A_1} - 1)\frac{z}{l}) \; + \; C_{i,L}}$$

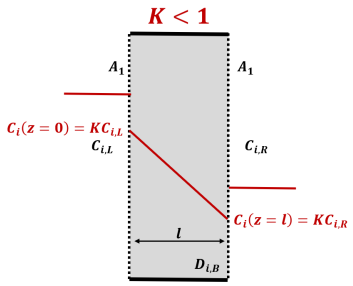
No longer is the concentration profile of $i$ a straight line. We know that the total amount of $i$ moving at all points of the membrane must be constant. As the cross-sectional area decreases left-to-right, $N_{i,z}$ must increase to maintain constant transport of $i$. The result is a concentration gradient that gets steeper (more downhill) as you move from left-to-right.



Again, consider the same physical scenario as part (a), except the membrane partitions $i$, such that $C_{i,membrane} = K \cdot C_{i,solution}$

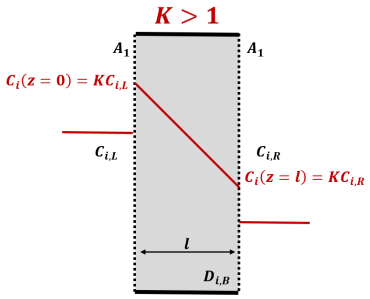
**c)** Sketch the concentration profile for $K < 1$. Are there discontinuities?

There is a Discontinuity in Concentration. At the membrane surface, $C_i$ is lower in the membrane than the fluid it is directly contacting.



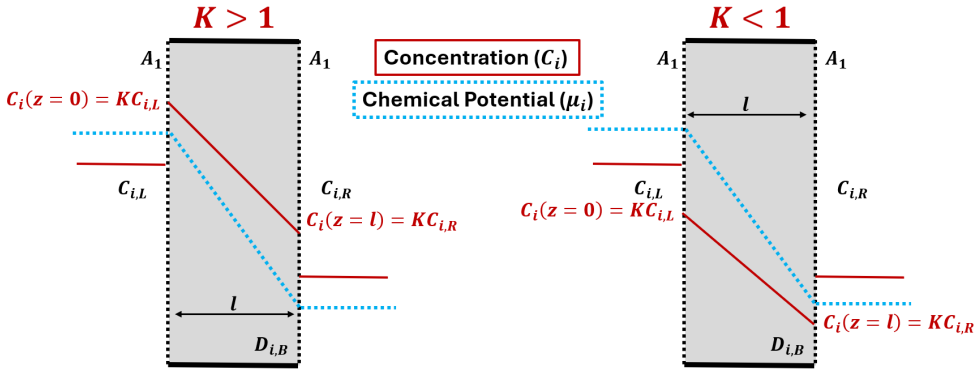

**d)** Sketch the concentration profile for $i$ when $K > 1$. Are there discontinuities?

There is a Discontinuity in Concentration. At the membrane surface, $C_i$ is higher in the membrane than the fluid it is directly contacting.




**e)** For the scenarios in parts (c) and (d), sketch the profile for chemical potential versus distance. Are there discontinuities? Why or why not?

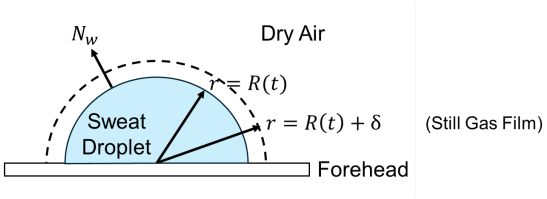
There can be no discontinuities in $\mu_i$. It would be aphysical. There is still a $\mu_i$ gradient across the membrane, which drives mass transport. The partition equation allows discontinuities in $C_i$ to persist at the membrane surface even though chemical potential does not.



## Problem 2 (x pts)

After finishing a marathon, you lay down in the grass to cool down. There is a hemispherical droplet of sweat on your forehead with radius $R = 0.5\ mm$. Assume the droplet on your forehead is parallel to the ground, the sweat droplet can be approximated as pure liquid water ($\rho_w = 1.00$ g cm$^{-3}$), and the surrounding air is dry ($y_\infty = 0$). Convection due to a gentle breeze establishes a still gas diffusion layer of constant thickness $\delta = 0.05$ mm surrounding the hemisphere such that $y_w(r = R + \delta) = 0$.

At 30 $°C$ and $P_{atm} = 1$ atm, the evaporation rate of the droplet is governed by diffusive transport through the stagnant gas film layer. $P_{sat,H_2O}(T = 30\ °C) = 0.042$ atm and $D_{H_2O,air}(30\ °C,\ 1\ \text{atm}) = 0.266\ cm^2\ s^{-1}$.




**a)** Perform a pseudo steady-state balance on water and air within the still film to calculate the maximum evaporation rate (mol min$^{-1}$).

First, determine whether we are in the dilute or concentrated regime.
Assume ideal gas: $y_w = \frac{P_w^{vap}}{P} = \frac{0.042\ \text{atm}}{1\ \text{atm}} = 0.042$ (dilute ✓)
Governing equation:
$\underline{\nabla} \cdot \underline{N}_A + \frac{\partial c_A}{\partial t} - R_A = 0 \rightarrow \underline{\nabla} \cdot \underline{N}_A = 0 = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 N_{A,r}) + \frac{1}{r sin\theta}\frac{\partial}{\partial \theta}(N_{A,\theta} sin\theta) + \frac{1}{r sin\theta}\frac{\partial}{\partial \phi}N_{A,\phi}$ Assume
$N_{A,\theta} = N_{A,\phi} = 0 \rightarrow 0 = \frac{\partial}{\partial r}(r^2 N_{A,r})$
$\underline{N}_A = -D_{AB}\underline{\nabla} x_A + x_A c_{tot}\underline{V} \rightarrow$ r component: $N_{A,r} = -D_{AB}c_{tot}\frac{\partial x_A}{\partial r} + x_A c_{tot} V_r$. For dilute limit ($x_A \approx 0$):
$N_{A,r} \approx -D_{AB}c_{tot}\frac{\partial x_A}{\partial r} \rightarrow 0 = \frac{\partial}{\partial r}(r^2[-D_{AB}c_{tot}\frac{\partial x_A}{\partial r}])$. Because $D_{AB}$, $c_{tot} \neq f(r)$, $0 = -D_{AB}c_{tot}\frac{\partial}{\partial r}(r^2\frac{\partial x_A}{\partial r})$
Simplifying, $0 = \frac{\partial}{\partial r}(r^2\frac{\partial x_A}{\partial r}) \rightarrow$ integrate, $C_1 = r^2\frac{\partial x_A}{\partial r}$.
Integrate again: $x_A = \frac{-C_1}{r} + C_2$.
BCs: $x_A(r = R + \delta) = 0,\ x_A(r = R) = x_{A0}$. Plugging in and solving yields $C_1 = \frac{x_{A0}R(R+\delta)}{\delta}$ and $C_2 = \frac{x_{A0}R}{\delta}$
so $\boldsymbol{x_A = \frac{x_{A0}R}{\delta}(1 - \frac{R+\delta}{r})}$.

We now need the general expression for flux. We will differentiate our concentration profile to yield flux ($f(r, R(t))$).
$N_{A,r} = -D_{AB}c_{tot}\frac{\partial}{\partial r}(\frac{-x_{A0}R(R+\delta)}{\delta}\frac{1}{r}) = \boldsymbol{\frac{-D_{AB}c_{tot}x_{A0}R(R+\delta)}{\delta}(\frac{1}{r^2})}$

To get evaporation rate (the positive), perform a mass balance about droplet (accounts for changing $R(t)$):
Take $m_A$ to be the total rate of mass evaporation from the droplet and $n_A$ to be the total rate of water evaporating from the droplet.
$\frac{dm_A}{dt} = -N_A|_R 2\pi R^2 MW_A$ (want positive evaporation rate)
Evaporation rate (mol time$^{-1}$)= $\frac{\text{mass}}{\text{time}} \cdot \frac{1}{MW}$
$\frac{dn_A}{dt} = \frac{dm_A}{dt} \cdot \frac{1}{MW_A} = \frac{1}{MW_A} \cdot 2\pi R^2 MW_A[\frac{D_{AB}c_{tot}x_{A0}(R+\delta)}{\delta R}] = \boldsymbol{\frac{2\pi D_{AB}c_{tot}x_{A0}R(R+\delta)}{\delta}} = \boldsymbol{\frac{dn_A}{dt}}$.

Now plug in values, check units.
$c_{tot} = \frac{P}{RT} = \frac{(1)\ \text{atm mol K}}{(0.082057)\text{L atm}(30+273.15)\text{K}} = 0.0402\frac{\text{mol}}{\text{L}}$.

Solve for time 0. This is when there will be the largest evaporation rate (large drop loses more total moles than small drop).
$\frac{dn_A}{dt} = \frac{2\pi(0.266)(0.0402)(0.5)(0.55)}{(0.05)}\frac{\text{cm}^2\ \text{mol mm mm}}{\text{s L mm}}\frac{(10^2)\text{cm}(1)\text{L}(60)\text{s}}{(10^3)\text{mm}(10^3)\text{cm}^3(1)\text{min}} = \boxed{9.5 \cdot 10^{-5}\frac{\text{mol}}{\text{min}}}$

**b)** How long will it take for the sweat droplet to evaporate?

We need to integrate an expression for $\frac{dR}{dt}$ expression from 0.5 mm to 0 mm which we can derive by converting mass to volume from $\frac{dm_A}{dt}$.

$$m_A = \rho_A V = \rho_A \cdot \tfrac{2}{3}\pi R^3$$

$$\frac{dm_A}{dt} = \frac{d(\rho_A \frac{2}{3}\pi R^3)}{dt} = \tfrac{2}{3}\pi\rho_A \cdot 3R^2\frac{dR}{dt} = 2\pi\rho_A R^2\frac{dR}{dt}$$

$$= \frac{-4\pi MW_A D_{AB} c_{tot} x_{A0} R(R+\delta)}{\delta} \cdot \frac{dR}{dt} = \frac{-2MW_A \cdot D_{AB} c_{tot} x_{A0}}{\delta\rho_A}\frac{(R+\delta)}{R} \rightarrow \boldsymbol{\frac{-2MW_A \cdot D_{AB} c_{tot} x_{A0}}{\delta\rho_A}\, dt = \frac{R}{R+\delta}\, dR.}$$

Integrating from 0 to $t_{evap}$ and $R_{initial}$ to 0:

$$t|_0^{t_{evap}} = \tfrac{1}{K}[(R+\delta) - \delta ln(R+\delta)|_{R_{initial}}^0], \text{ where } K = -0.032336\tfrac{\text{mm}}{\text{s}}$$

Units for $K$: $\frac{\text{g cm}^2 \text{ mol cm}^3}{\text{mol s L mm g}} = \frac{(\text{cm}^5)\cdot\ (1)\ \text{L}\ (10)\ \text{mm}\ (10)\ \text{mm}}{(\text{s L mm})\cdot(10^3)\ \text{cm}^3\ (1)\ \text{cm}\ (1)\ \text{cm}} = \frac{1}{10}\text{mm/s}$

This will give $t_{evap}$ units of seconds for $R, \delta[=]$ mm (min or sec accepted). The integral was solved with u-substitution $u = R + \delta$. Write out the unit conversions to get the correct units for $K$!

$$t_{evap} = \tfrac{1}{K}[\delta ln(\tfrac{R+\delta}{\delta}) - R_{initial}] = \boxed{23 \text{ seconds}}.$$

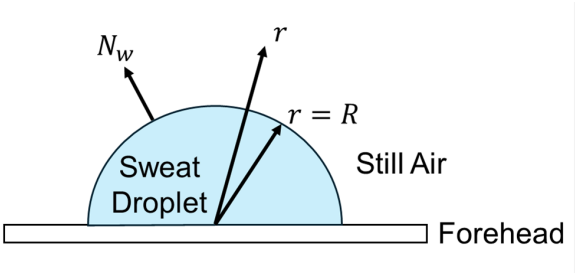
This order of magnitude makes sense for the evaporation of a tiny bead of sweat!

**c)** The latent heat of vaporization of water is 2.26 kJ $g^{-1}$. What is the rate of heat removal due to the evaporation of the sweat hemisphere (in kW)?

$$\frac{(9.5E{-}5)\ \text{mol}}{\text{min}}\frac{(2.26)\ \text{kJ}}{\text{g}}\frac{(18)\ \text{g}}{\text{mol}}\frac{(1)\ \text{min}}{(60)\ \text{s}} = \boxed{6.5 \cdot 10^{-5}\ \text{kW}}$$

**d)** Now suppose that $\delta << R$. Sketch the simplified geometry of the problem and perform a pseudo steady-state balance on the water and the air to derive the concentration profile for water, $y_w$, within the film. Sketch this concentration profile.

This is a quick reinforcement of basic concentration profiles/considering problem geometries.




Updated BCs: $x_A(z = R) = x_{A0}$, $x_A(z = R + \delta) = 0$.

Same $\underline{\nabla} \cdot \underline{N}_A = 0$, but in Cartesian 1D now:

Simplifies to $0 = \frac{dN_{A,z}}{dz}$.

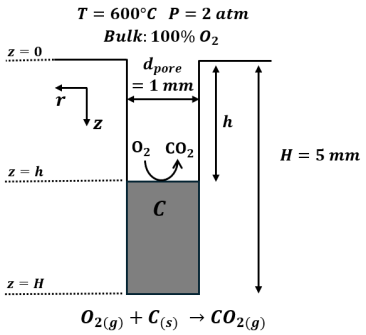
Integration yields: $x_A = C_1 z + C_2$. Plug in BCs $\rightarrow x_A = \frac{x_{A0}}{\delta}(z - (R - \delta))$

$$\boxed{y_w = 0.84(z - .45)}$$

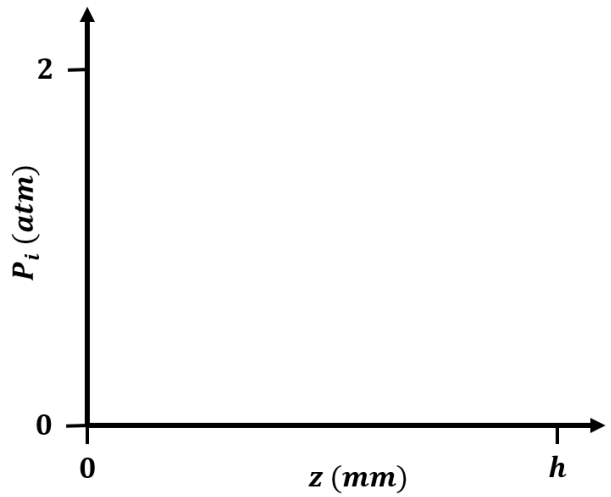
+ Linear profile with correct BCs

## Problem 3 (X pts)

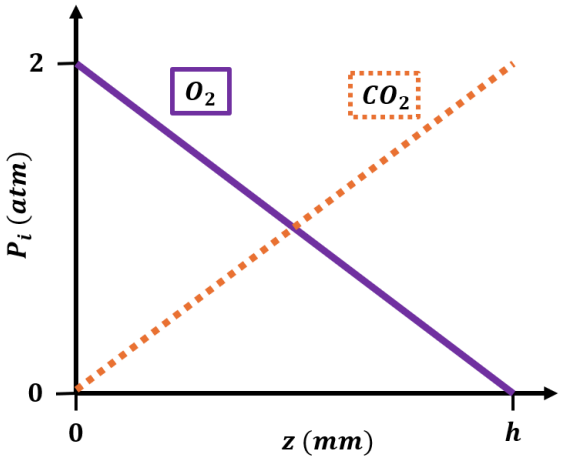
Fluidized catalytic cracking (FCC) is an industrial refining process that turns heavy hydrocarbon mixtures into lighter hydrocarbons for making fuels. During, use the catalysts used for FCC processes become covered in carbonaceous "coke" which must be removed. Pure oxygen gas ($O_2$) at high temperatures is used to oxidize the coke (solid $C$) to carbon dioxide gas ($CO_2$). This process removes the solid carbon clogging catalyst pores, enabling catalyst reuse. You may assume that excess $O_2$ is used, allowing the bulk gas to remain pure $O_2$. You may also assume that the rate of reaction between $C$ and $O_2$ is rapid compared to the rate of diffusion, leading to mass-transport limited $CO_2$ production and near-zero $O_2$ concentrations at the coke surface. The catalyst of interest has pores which are cylindrical, with a diameter of 1 $mm$ and depth of 5 $mm$. The oxidative cleaning process is carried out at 600 $°C$ and 2 $atm$. The density of solid carbon $\rho_C$ is 2.25 $\frac{g}{cm^3}$. You may assume the Hirschfeld corelation is appropriate for determining diffusivities.




**a)** Sketch the concentration profile you expect to see for $O_2$ and $CO_2$ some time after beginning the oxidative cleaning?




The steady-state profiles of $O_2$ and $CO_2$ can be seen on the figure below. We know that at the top of the pore, where $z = 0$, the gas phase is entirely $O_2$. We also know that the rate of $C$ combustion is rapid enough to make the concentration of $O_2$ at $z = h$ equal to zero. These boundary conditions and the profile below are true for the system at any $t > 0$ and any value of $h > 0$




**b)** Starting from a material balance, derive an expression for the pseudo-steady-state flux of $O_2$ through the pore.

We will begin solving for PSSH flux of $O_2$ by using a generic mass balance.
$In \ - Out \ + Generation = Accumulation$

Applying this material balance as a shell balance with 1-D flux in the z direction:
$\frac{d(N_{O_2} \cdot A)}{dz}|_z - \frac{d(N_{O_2} \cdot A)}{dz}|_{z+\Delta z} + R_{O_2} = \frac{dC_{O_2}}{dt}$
How can we simplify the material balance?:
Pseude-Steady State assumption: $Accumulateion = \frac{dC_{O_2}}{dt} = 0$
No Bulk Reaction of $O_2$: $Generation = R_{O_2} = 0$

Simplified Material Balance:
$\frac{d(N_{O_2} \cdot A)}{dz}|_z - \frac{d(N_{O_2} \cdot A)}{dz}|_{z+\Delta z} = 0$
Factor A out of the derivative (permissible because cross-sectional area of the pore is constant), divide both sides by $\Delta z$, and apply the definition of a derivative:
$\frac{dN_{O_2}}{dz} = 0$

Thus, we know that $N_{O_2}$ is constant.

Now, let's look at the expression for flux: $N_{O_2,z} = -c_{tot} D_{O_2,CO_2} \frac{dy_{O_2}}{dz} + y_{O_2}(N_{O2,z} + N_{CO_2,z})$

In this scenario, we can apply the condition of equimolar counter-diffusion for $O_2$ and $CO_2$ in the pore:
$N_{O_2,z} = -N_{CO_2,z}$, meaning $(N_{O2,z} + N_{CO_2,z}) = 0$
This allows us to simplify the expression for flux:
$N_{O_2,z} = -c_{tot} D_{O_2,CO_2} \frac{dy_{O_2}}{dz}$

Separating, we find:
$\frac{-N_{O_2}}{c_{tot} D_{O_2,CO_2}} dz = dy_{O_2}$

Integrating over appropriate bounds:
$\frac{-N_{O_2}}{C_{tot} D_{O_2,CO_2}} \int_0^h dz = \int_1^0 dy_{O_2}$

$\frac{-N_{O_2}}{c_{tot} D_{O_2,CO_2}} (h - 0) = 0 - 1$

$$\boxed{N_{O_2} = \frac{c_{tot} D_{O_2,CO_2}}{h}}$$

**c)** Some time after starting the oxidation process, the pore has been cleared to a depth of 3.0 $mm$. What is the production rate of $CO_2$ at this point of the process?

To complete this problem, you must recognize that every molecule of $CO_2$ produced in the combustion reaction at $z = h$ must be equal to $N_{CO_2}$ moving away from $z = h$. Subsequently this means that
$|N_{O_2}| = |N_{CO_2}| = CO_2\ Production\ Rate$
Thus, to calculate $CO_2$ production rate, you need to calculate $D_{O_2,CO_2}$ and $c_{tot}$.

Calculating $c_tot$

$$c_{tot} = \frac{n_{tot}}{V} = \frac{P}{RT} = \frac{2\ atm}{(873K)(0.08206\frac{L\cdot atm}{mol\cdot K})} = 0.0279\ M$$

Calculating $D_{O_2,CO_2}$ Using Hirschfelder:
Relevant Physical Properties (Appendix K):
$CO_2$: $\sigma_{CO_2} = 3.996\ \mathring{A}$, $\frac{\epsilon_{CO_2}}{k} = 190\ K$, $M_{CO_2} = 44.01\frac{g}{mol}$
$O_2$: $\sigma_{O_2} = 3.433\ \mathring{A}$, $\frac{\epsilon_{O_2}}{k} = 113\ K$, $M_{O_2} = 32.0\frac{g}{mol}$

Calculated Hirschfeld Parameters

$$\sigma_{O_2,CO_2} = 3.715\ \mathring{A},\ \frac{\epsilon_{O_2,CO_2}}{k} = 146.5\ K,\ \boxed{\Omega_{D,O_2,CO_2}(T = 873\ K) = 0.814},\ P = 2\ atm,\ T = 600\ °C = 873\ K$$

Hirschfeld Diffusivity Calculation

$$D_{O_2,CO_2} = \frac{0.001858(873\ K)^{3/2}(\frac{1}{44.01} + \frac{1}{32.0})^{1/2}}{(2\ atm)(3.715\mathring{A})^2(.814)} = \boxed{0.495\ \frac{cm^2}{s}}$$

Now, Calculate the Flux of $O_2$ when $h = 0.3\ cm$:

$$N_{O_2} = \frac{c_{tot}D_{O_2,CO_2}}{h} = \frac{(0.0279\frac{mol}{L})(0.495\frac{cm^2}{s})(.001\frac{L}{cm^3})}{0.3\ cm} = 4.60\times10^{-5}\ \frac{mol}{cm^2\cdot s}$$

Now, determine the total amount of $O_2$ transfer, and thus $CO_2$ production, by multiplying by the cross sectional area of the pore

$$r_{CO_2} = N_{O_2}\cdot A_{pore} = 4.60\times10^{-5}\ \frac{mol}{cm^2\cdot s}\times\frac{\pi(0.1cm)^2}{4} = \boxed{3.616\times10^{-7}\frac{mol\ CO_2}{s}}$$

**d)** How long will the cleaning process take to clean a depth of 4.0 $mm$ from the mouth of the pore?

To know how long a pore will take to clear to 4.0 mm, we must calculate a function for $h(t)$. This means we must do a mass balance around the carbon.

$$\frac{dm_C}{dt} = N_{O_2}\cdot S\cdot MW_C$$

Plug in the previous expression for $N_{O_2}$ and the expression $m_c = h\cdot S\cdot\rho_C$:

$$\rho_C\frac{dh}{dt} = \frac{c_{tot}D_{O_2,CO_2}}{h}\cdot MW_C$$

Separate the derivative:

$$hdh = \frac{c_{tot}MW_CD_{O_2,CO_2}}{\rho_C}dt$$

Integrate once:

$$\frac{h^2}{2} = \frac{c_{tot}MW_CD_{O_2,CO_2}}{\rho_C}t + C_3$$

Apply Initial Condition $h(t = 0) = 0$:
$0 = 0 + C_3$ and therefore $C_3 = 0$

After simplifying the expression with the knowledge that $C_3 = 0$:

$$h(t) = \sqrt{\frac{2c_{tot}MW_CD_{O_2,CO_2}}{\rho_C}t}$$

To solve the for the time required to clear 4 mm of the pore, you must simplify the expression above to solve for $t$:

$$t = \frac{h^2 \rho_C}{2 c_{tot} M W_C D_{O_2, CO_2}}$$

Solving for $h = 0.4\ cm$:

$$t(h = 4\ mm) = \frac{(0.4\ cm)^2 (2.25\ \frac{g}{cm^3})}{2(0.0279 M \times .001 \frac{L}{cm^3})(12 \frac{g}{mol})(0.495 \frac{cm^2}{s})} = \boxed{1086\ seconds = 18.1\ minutes}$$

**e)** When is the rate of $CO_2$ emission the highest. Justify your answer using physical reasoning.

$r_{CO_2}$ will be highest at the moment instantly after $t = 0$, when $h = 0$. This is because $O_2$ will have an infinitesimally small distance to diffuse. As soon as the pore has any depth where $h > 0$, the development of a concentration gradient in oxygen will slow down $O_2$ flux, and the resulting process of coke combustion.

## Problem 4 (X pts)

Tropical carnivorous "pitcher plants" lure their prey into their leaves with a sweet-smelling 1-hexanol. 1-hexanol sits in a reservoir at the base of their pitcher-shaped leaves and diffuses up before being swept away by the surrounding air. This system can be approximated by a cylindrical Arnold Cell of radius $R = 5\ cm$, and height $H = 25\ cm$ with air flowing perpendicular to the mouth of the cell at a velocity $v$ (see below). The plants grow at a temperature of 25 $°C$ and atmospheric pressure ($P_{atm} = 1$ atm).

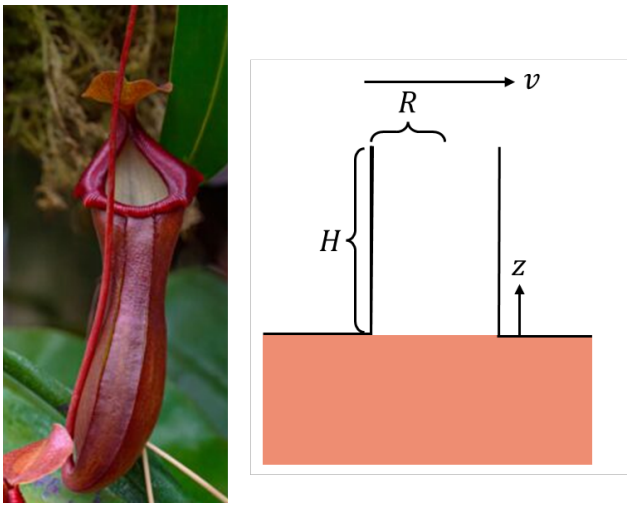



Figure 1: Image of tropical pitcher plant leaf (left) and approximated Arnold Cell (right).

The vapor pressure of 1-hexanol can be approximated with Antoine's equation of the following form:

$$log_{10}(P_{sat}) = A - \frac{B}{T + C},$$

where $A = 4.413$, $B = 1422$, $C = -107.7$, $T[=]K$, and $P[=]$ bar (source: NIST).

The density of 1-hexanol is 0.85 $g\ cm^{-3}$, molecular weight is 102.17 $g\ mol^{-1}$, and diffusivity in air is 6.21 $\cdot 10^{-6}\ m^2\ s^{-1}$ (source: Lugg, G.A.: Anal. Chem. 40 (1968) 1072–1077).

**a)** Calculate the vapor fraction of 1-hexanol, $y_h$ just above the surface of the 1-hexanol reservoir. Without solving, plot (in Python, Excel, or other computational tool) the expected concentration profile of 1-hexanol ($y_h$ vs. $z$) for $T = 25\ °C$.

$P_{sat}(25°\ C) = 10^{(4.413 - \frac{1422}{293 - 107.7})} = 0.000884$ atm. Assuming ideal gas,
$y_h(z = 0) = \frac{0.000884\ \text{atm}}{1\ \text{atm}} = \boxed{0.000884}$ (Note: dilute!)
$\boxed{+\ \text{Plot (linear again!) from (0,0.000884) to (25, 0)}}$

Now suppose the pitcher plant grows on a planet with an identical atmosphere to Earth, but with a temperature of 150 $°C$ ($P_{atm} = 1$ atm). Assume the diffusivity and density of 1-hexanol are constant, and the given Antoine's relation can be extrapolated to this temperature.

**b)** Recalculate $y_h(z = 0)$ and perform steady-state balances on 1-hexanol and air within the pitcher cavity to calculate the molar flux of 1-hexanol, $N_h$, when $T = 150\ °C$.

Using Antoine's again, $P_{sat}(T = 150\ °C) = 0.8038$ atm, $\boxed{y_h = 0.8038}$ (Note: now concentrated!)
$\underline{\nabla} \cdot \underline{N}_h + \frac{\partial c_h}{\partial t} - R_A = 0.$
Steady state, no reaction, 1D rectilinear: $0 = \frac{dN_{h,z}}{dz} = \frac{dN_{air,z}}{dz}$
BCs: $y_h(z = 0) = y_{h0} = 0.8038$, $y_h(z = H) = 0$ Flux expression:
$\underline{N}_h = -D_{h,air} c_{tot} \underline{\nabla} y_h + y_h(\underline{N}_h + \underline{N_{air}})$

Assume air is insoluble in 1-hexanol: $N_{air} = 0$.

$\rightarrow N_{h,z} = -D_{h,air}c_{tot}\frac{dy_h}{dz} + y_h N_{h,z}$ Rearrange, $N_{h,z} = \frac{-D_{h,air}c_{tot}}{1-y_h}\frac{dy_h}{dz}$

Because $N_{h,z} \neq f(z, y_h)$ for this 1D problem, we can solve directly from here. Separate and integrate as follows.

$$\frac{-N_{h,z}}{D_{h,air}c_{tot}}\int dz = \int \frac{1}{1-y_h}dy_h \rightarrow \frac{N_{h,z}}{D_{h,air}c_{tot}}z = -ln(1-y_h) + C$$

In this case, we can solve directly for $C$ with our concentration boundary conditions at the surface ($y_h(z=0) = y_{h0}$) and top of the Arnold Cell ($y_h(z=H) = 0$).

$$0 = -ln(1-y_{h0}) + C \rightarrow C = ln(1-y_{h0}) \rightarrow N_{h,z} = \frac{-D_{h,air}c_{tot}}{z}ln(\frac{1-y_{h0}}{1-y_h}).$$

Plugging in the second BC at the top of the Arnold Cell ($y_h(z=H)=0$): $N_{h,z} = \frac{-D_{h,air}c_{tot}}{H}ln(1-y_{h0})$

$$c_{tot} = N/V = \frac{1 \text{ atm mol K}}{0.082057 \text{ L atm}(150+273.15) \text{ K}} = 0.0287999 \text{ mol/L}.$$

Pay attention to units again! Remember to convert diffusivity, concentration, and height to the same unit (I used cm).

$$-7.154 \cdot 10^{-8} \cdot ln(1-0.8038)\frac{\text{mol}}{\text{cm}^2\text{s}} \quad \boxed{N_{h,z} = 4.39 \cdot 10^{-8}\frac{\text{mol}}{\text{cm}^2\text{s}}}$$

**c)** On the same axes as (a), plot the concentration profiles, $y_h$ and $y_{air}$ vs. $z$ at $T = 150\ ^\circ C$.

You can solve all the way from the general balance by plugging in the flux expression and integrating twice to get concentration. Since we already have the flux expression, though (rectilinear coordinates), we can solve directly from our flux expression.

$$N_{h,z} = \frac{-D_{h,air}c_{tot}}{z}ln(\frac{1-y_{h0}}{1-y_h})$$

$$\exp(\frac{zN_{h,z}}{D_{h,air}c_{tot}}) = \frac{1-y_h}{1-y_{h0}} \quad y_h = 1-(1-y_{h0})\cdot \exp(\frac{zN_{h,z}}{D_{h,air}c_{tot}}) =$$

$$1-(1-y_{h0})\cdot \exp(\frac{z}{D_{h,air}c_{tot}}\frac{-D_{h,air}c_{tot}ln(1-y_{h0})}{H}) = 1-(1-y_{h0})\cdot \exp(\frac{-z}{H}ln(1-y_{h0}))$$

$$\boxed{y_h = 1-(1-y_{h0})\cdot \exp(\frac{-z}{H}ln(1-y_{h0}))}$$

Plot on same axes (same endpoints), but concave down in the middle

Note: diffusivity and total concentration do not appear in any of our concentration profiles! Does this make sense to you?

**d)** What is the net flux of air, $N_{air}$ in the pitcher cavity? Given the profile of $y_{air}$ at the elevated temperature, what balancing modes of transport maintain $N_{air}$? How is this different physically from the low temperature case?

The net flux of air in the cavity is $\boxed{0}$. Mathematically this appears as $\frac{dN_{air}}{dz} = 0$. $N_{air}(z=0) = 0$ because air is assumed to be insoluble in the liquid.

At the bottom of the Arnold cell (high temperature case):

- The concentration of 1-hexanol is high and air is low.
- 1-hexanol moves up by diffusion (down concentration gradient) and by convection
- Air moves down by diffusion (down concentration gradient) and up by convection (carried by 1-hexanol) to maintain a net 0 flux.

At the top of the Arnold cell (high temperature case):

- The concentration of 1-hexanol is low and air is high
- 1-hexanol moves up by diffusion (down concentration gradient), but low concentration means convection negligible (think about our expression for $N_h$!
- Air moves down via diffusion and up via convection (equal and opposite to maintain net 0 flux of air)